COLIN NGUYEN

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AI engineer building production LLM agents and the evaluation infrastructure that keeps them reliable in regulated fintech.

EXPERIENCE

Athena Money *Nov 2025 – Present*
Founding AI Engineer *San Francisco Bay Area*

- Built the team's multi-turn LLM evaluation platform for regulated mortgage underwriting (scenario graders, LLM-as-judge rubrics, per-turn scoring, an LLM user simulator); cut the supervisor evaluation run from about 55 minutes to 15.
- Took run-to-run noise out of agent evaluation: field-level consensus across trials, confidence calibration and regression gates in tiered CI on Temporal; an audit found eight framework and grader bugs, fixed in five days.
- Shipped production underwriting and follow-up agents (Google ADK, Temporal) that validate every action in the turn and fail closed under a supervising layer and a borrower override; retired the legacy workflow behind flags.
- Delivered the joint-application lifecycle end to end across backend, web and both mobile apps: co-borrower invites and proxies, joining at rate lock, per-applicant disclosures, identity mismatches routed to review.
- Brought the Android app (Kotlin, Jetpack Compose) from a partial shell to iOS parity and cut its CI critical path from 26 to 6 minutes; put 478 step tests and 45 iOS screens under blocking snapshot tests, from none.

Zeta Global *Jun 2025 – Aug 2025*
Data Science & Generative AI Intern *San Francisco*

- Cut manual workflow setup by 70% with a LangGraph supervisor agent that routes tasks to specialised sub-agents.
- Built a multi-vendor (OpenAI, Anthropic, Bedrock) orchestration platform on FastAPI with PostgreSQL checkpointing and an evaluation framework (accuracy, consistency, latency, cost); deployed via REST APIs on Docker and Kubernetes.

NationBuilder *Jan 2025 – Nov 2025*
AI Research Intern (UCSB capstone project through Jun 2025) *Remote*

- Ranked document relevance at 93% classification accuracy with a RAG pipeline (all-MiniLM-L6-v2 embeddings, FAISS), so irrelevant PDFs are skipped before processing.

RESEARCH

Risk-Controlled Decision Gates — with Prof. Eric Suess, CSU East Bay *Oct 2026 – Present*

- Showed a Learn-then-Test abstention guarantee (5% risk) fails under option-count shift: 7.2% realised risk from 5 to 150 options on a public decision model, 7.7% on SmolLM2 models tuned to match it; pre-registered.

Efficient LLM Fine-Tuning — with Prof. Eric Suess, CSU East Bay *Sep 2026 – Oct 2026*

- Raised SmolLM2's trainable step sixteenfold (8,192 to 131,072 tokens) in 71% of baseline memory with five stacked techniques; hand-wrote a block-wise 8-bit AdamW matching torchao byte for byte and fp32 across six seeds.

Evaluation of Mobile-Health Receptivity Models — with R. Benzo (OSU) and P. Washington (UCSF) *Jun 2026 – Present*

- Showed that the standard pooled metric for receptivity models is dominated by between-person comparisons across three cohorts, so it mostly rewards knowing who tends to respond; pre-registered personalisation study; manuscript in preparation.

SELECTED PROJECTS

Pokémon TCG AI Battle Challenge — Kaggle *Aug 2026 – Sep 2026*

- Placed 2,166th of 6,807 on a live ladder with a hidden-information card-game agent (scripted policy, determinized whole-turn search, self-play value network); then audited the claims and kept only what the logs support.

NBA Catch Decisions — independent, on public player-tracking data *Oct 2026*

- Built a catch-level dataset from the only public NBA tracking data (621 games, 2015–16) and showed players trade a foot of space at the catch for about eight seconds of shot clock; every test pre-registered, failures kept.

Paper Trader — Alpaca paper account *Aug 2026 – Present*

- Built an ETF-trading system that commits pass/fail criteria to git before any backtest and checks each live decision against a simulated twin; 26 of 27 matched, and it failed two of its own reliability criteria.

SKILLS

Languages: Python, Go, Kotlin, Swift, SQL, Java, JavaScript, R, C
LLM / AI: LangGraph, Google ADK, Temporal, MCP and tool use, RAG (FAISS), LLM-as-judge evaluation, Claude Code
ML: PyTorch, PEFT (LoRA, QLoRA), torchao, vLLM, scikit-learn, TensorFlow, conformal risk control, survival analysis
Infra / MLOps: AWS, Google Cloud, Docker, Kubernetes, FastAPI, PostgreSQL, GitHub Actions, Paparazzi, OAuth 2.0, Spark

EDUCATION AND PATENTS

University of California, Santa Barbara *Sep 2022 – Jun 2025*
B.S. in Statistics & Data Science, Minor in Mathematics *Santa Barbara, CA*

- "Adaptive Neural Content Generation", U.S. Provisional App. No. 63/943,780, filed Dec 2025; first-named inventor.